

OLIVER WILDE

MCOMP GAME ENGINEER STUDENT

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A passionate programmer that feels way too much enjoyment from drawing triangles on a screen.

EDUCATION

Newcastle University, United Kingdom

Integrated Masters in Computer Games Engineering

September 2021 - Present

Overall: TBD - On track for 1st Class

Holy Cross College, Manchester, United Kingdom

September 2018 - Sept 2020

A-Levels: Chemistry, Biology, Psychology, EPQ

Blackpool Aspire Academy

September 2016 - Sept 2018

GCSEs: 98888877

NOTABLE PROJECTS

An extensive description of all of my projects can be found on my website, which is being continuously updated with my new works

Vulkan Voxel Rendering Engine (Ongoing)

- Ongoing dissertation project focusing on real-time voxel terrain in Vulkan. It is more focused on researching how to minimise memory usage, optimising update overhead, and sustaining stable frame rates. Written with **C++** with **GLFW**, **GLM**, and **Vulkan**

Physics-Focused Unity FPS Shooter

- Created a first-person shooter in Unity leveraging **C#** to emphasize realistic physics, robust player movement, and dynamic world interactions. Implemented advanced collision volumes, dynamic gravity zones, and a day-night cycle, while structuring enemies with basic state-machine AI and NavMesh pathfinding. Despite simpler enemy behavior, the project excelled at showcasing nuanced physics responses, scene transitions (Main Menu / Game / Game Over), and efficient prefab use for weapons, props, and interactive objects.

Game Design Document - Shifting Sands: Outpost Defense

- Created a first-person shooter in Unity leveraging **C#** to emphasize realistic physics, robust player movement, and dynamic world interactions. Implemented advanced collision volumes, dynamic gravity zones, and a day-night cycle, while structuring enemies with basic state-machine AI and NavMesh pathfinding. Despite simpler enemy behavior, the project excelled at showcasing nuanced physics responses, scene transitions (Main Menu / Game / Game Over), and efficient prefab use for weapons, props, and interactive objects.

Team Fitness Companion App (React Native & Firebase)

- Collaboratively developed a cross-platform fitness tracking app featuring user authentication, data storage, and a responsive UI. Leveraged React Native for front-end architecture, Firebase for back-end integration, and Git/GitHub for version control. Key contributions included implementing login and admin control pages, establishing best-practice code organization, and streamlining team onboarding by resolving environment and dependency issues. Enhanced skills in React Native, Git workflows, time management, and team communication to deliver a cohesive, user-friendly product.

Implementing and Evaluating a CNN on CIFAR-10

- Developed a multi-layer Convolutional Neural Network in Python (using PyTorch, NumPy, and Matplotlib) for CIFAR-10 image classification. The project demonstrated effective early stopping to preserve optimal weights, showcased improved training stability through Batch Normalization, and included visualization of learned filters and feature maps to illustrate progressive feature abstraction.

Auction House (University Team Project)

- Delivered a working console-driven auction platform, emphasizing teamwork and database design.

Personal Website (Ongoing)

- Developed a React Website that has a more extensive review of my best work, continuously adding more projects and features as development continues. Written with **HTML**, **CSS**, **Javascript** within the **React framework**.

Wordle Game Enhancements

- Second-year project improving the classic Wordle game with advanced logic for repeated letters, a strategic single-letter hint system, and a Hard Mode enforcing previously revealed clues—all built with TDD to ensure reliability. Developed in Python with a focus on clear console-based user interface.

TECHNICAL SKILLS

Programming Languages	C++, C#, Java, C, Python, HTML, CSS, Javascript
Tools:	Visual Studio, VS Code, Git, Cmake, IntelliJ, PyCharm, DBeaver, Figma, Notion
Frameworks, Libraries, and Engines:	Unity, React/ React Native, Vulkan, NumPy, Matplotlib

WORK EXPERIENCE

Bartender at Digital Newcastle (Tokyo Industries)	Jan 2023 - April 2024
Worked in a high volume bar environment serving drinks whilst upholding great customer service and satisfaction.	
Dental Intern (Observer) – Synergy Dental Clinic	Feb 2020 – Mar 2020
- Shadowed a dentist for 3-4 weeks, observing patient care and procedures. - Strengthened interpersonal communication and attention to detail.	

PERSONAL NOTES

I love everything related to electronics, so I have picked up the hobby of bedroom production with my **Arturia Minilab 3**.
Enjoys boxing, snooker, walking in nature, and video games.

REFERENCES

Available upon Request. Please contact me via email or phone directly.